

Optimizing field service operations with NLP and deep learning-based spare parts prediction

Alessio Schiavo^{a, b} , Alessandro Renda^{c, *} , Pietro Ducange^a , Francesco Marcelloni^a 

^a Department of Information Engineering, University of Pisa, Largo Lucio Lazzarino 1, Pisa, 56122, Italy

^b LogObject AG, Thurgauerstrasse 101a, Opfikon, 8152, Switzerland

^c Department of Engineering and Architecture, University of Trieste, Via Valerio 6/1, Trieste, 34127, Italy

HIGHLIGHTS

- Novel approach based on NLP and deep learning for spare parts prediction in appliance repair.
- Uses structured data and textual fault descriptions to enhance predictive accuracy.
- Outperforms state-of-the-art methods in two real-world industrial cases.
- Cuts unnecessary field service technician visits by up to 64%, improving service efficiency.
- Quality of textual fault descriptions is key to predictive performance.

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ABSTRACT

Field service technicians typically rely on their experience and fault descriptions to identify and select the appropriate spare parts for repairing an appliance. However, missing parts often lead to multiple service visits, increasing operational costs and reducing customer satisfaction. To address this issue, we propose DL-SPP.*sbert*, a Deep Learning-based Spare Part Prediction system that leverages NLP and structured data to accurately predict the spare parts needed for repairs. DL-SPP.*sbert* processes historical repair data, appliance metadata, and fault descriptions by leveraging SBERT sentence embeddings to capture the semantic meaning of textual inputs. Unlike retrieval-based methods, our system directly learns the relationships between input features and spare parts usage through a multi-label classification model. DL-SPP.*sbert* was evaluated on two real-world case studies, where it achieved up to a 64 % reduction in multiple-visit interventions, surpassing the performance of state-of-the-art baselines. A detailed explainability study highlights the key role of fault descriptions and appliance metadata in prediction accuracy, confirming DL-SPP.*sbert* as an effective tool for optimizing spare parts provisioning. Finally, experimental results indicate that updating the model with newly collected data is fundamental to preserving predictive performance over time in continuously evolving real-world industrial settings.

1. Introduction

Optimizing field service operations is essential for companies that offer repair and maintenance services to customers. When an appliance malfunctions, resolving the issue quickly and effectively is essential to maintaining customer satisfaction. Ideally, repairs should be completed in a single on-site visit by the Field Service Technician (FST). Each additional visit increases operational costs, including labor and travel expenses, and negatively impacts the customer experience. As a result,

companies aim to minimize repeat visits to boost both efficiency and service quality.

While extensive research has been conducted on inventory management optimization [1] and spare parts demand forecasting [2], relatively little attention has been devoted to automated solutions that can predict the specific spare parts needed for individual repair interventions. Traditionally, this decision relies on the expertise of FSTs, introducing variability and reducing operational efficiency. Furthermore, the knowledge is concentrated in the hands of the FST, creating a potential

* Corresponding author.

Email address: alessandro.renda@dia.units.it (A. Renda).

risk in the event of their departure from the company. The increasing availability of digital maintenance records and historical repair data presents a strategic opportunity to move beyond manual, experience-driven decisions and adopt predictive, data-informed methodologies. However, effectively leveraging this information is challenging due to the unstructured nature of textual fault reports, which limits the applicability of conventional predictive modeling techniques in this domain.

In a previous work [3], we introduced a Case-Based Reasoning (CBR)-driven Spare Parts Prediction system (CBR-SPP), which sets a new state of the art in spare parts prediction by leveraging historical repair cases. The CBR-SPP system maintains a structured case base of past repair interventions, including textual fault descriptions, appliance metadata, and the corresponding spare parts used. Given a new repair request, the system processes the fault description using Natural Language Processing (NLP) techniques to retrieve relevant past cases and rank spare parts based on their relevance.

While the CBR-SPP system has proven effective in reducing unnecessary FST visits, its dependence on structured retrieval methods introduces inherent limitations. The search process prioritizes cases matching the same appliance type before evaluating textual similarity in fault descriptions. While this approach enhances interpretability, it narrows the search space, limiting the system's ability to generalize to unseen cases or underrepresented appliance models. Moreover, as the case base expands, retrieval times increase, resulting in higher computational overhead and reduced suitability for real-time inference.

In this paper, we propose Deep Learning-based Spare Parts Prediction (DL-SPP.*sbert*), a novel methodology that replaces structured case retrieval with a deep learning model, trained on past repair interventions. Like CBR-SPP, DL-SPP.*sbert* leverages NLP techniques to process textual fault descriptions, but instead of retrieving similar cases, it learns direct relationships between fault descriptions, appliance metadata, and spare parts usage through a Multi-Layer Perceptron (MLP). The MLP takes as input SBERT-derived semantic embeddings [4], along with structured categorical attributes of the appliance, and formulates the prediction task as a multi-label classification problem, where each label represents a spare part used in historical repair interventions.

By eliminating the explicit retrieval step, DL-SPP.*sbert* removes dependency on a manually maintained case base and significantly improves scalability. Instead of searching for similar past cases at inference time, the model learns to generalize patterns from historical data, enabling fast and efficient predictions even as the dataset grows. This makes DL-SPP.*sbert* particularly well-suited for large-scale, dynamic repair environments where new failure patterns continuously emerge.

To comprehensively evaluate DL-SPP.*sbert*, we benchmark it against three comparative approaches: CBR-SPP, as presented in [3], a simple frequency-based approach, and a traditional deep learning pipeline combining BERT-based token embeddings with 1D-CNN-based feature extraction. We used the same real-world case study introduced in [3] and a new dataset from another company to assess the generalization capabilities of both approaches. This expanded evaluation framework enables a thorough validation of the proposed method's robustness and scalability.

Beyond predictive accuracy, this work addresses two critical aspects for the deployment of data-driven decision-support systems in industrial environments: temporal robustness and model explainability. First, we analyze the impact of concept drift [5] on our SPP system, by studying how model performance evolves over time and by evaluating different learning strategies designed to adapt to changing data distributions. Second, we provide illustrative examples of post-hoc local explanations based on Shapley values [6], offering insights into the contribution of textual and structured features to individual predictions.

The rest of the paper is organized as follows. In Section 2, we provide an overview of related works and some preliminaries on DL and NLP techniques. In Section 3, we describe our method. In Section 4, we introduce two real-world case studies, outline the experimental setup, and

present the evaluation metrics adopted to assess the proposed solution. Furthermore, we introduce the baseline approaches considered in our comparative experimental analysis. In Section 5, we first present the quantitative results of the experimental comparison between DL-SPP and CBR-SPP, which serves as the baseline model. Then, we conduct an ablation study to evaluate the contribution of each feature to the predictive model from a global perspective, followed by a local explainability analysis on representative maintenance scenarios. Finally, we report the results of our analysis on model update strategies to address concept drift issues. In Section 6, we summarize our findings and draw concluding remarks.

2. Related works

The problem addressed in this study bears a strong resemblance to what is traditionally known in Operations Research as the *repair kit problem* [7]. This problem involves determining the optimal assortment of spare parts to be carried in service vans by FSTs, with the objective of minimizing both unsuccessful repair interventions and the costs associated with over-provisioning. Most of the recent literature [8–13] focuses on the multiple-job repair kit problem [14], where FSTs are expected to visit several customer sites before replenishing their stock. In these formulations, the problem is addressed as a combinatorial optimization task, where a key challenge is balancing holding costs (i.e., the value of the inventory carried) against penalty costs incurred when an essential part is unavailable during a visit. Given the vast number of possible spare part combinations and job configurations, exact solutions are computationally infeasible in realistic settings, prompting the use of heuristics. Notably, unlike our approach, most of these works treat the probability $p_{sp}^{t_j}$ —that a part sp is required for a task t_j —as a fixed input, typically estimated via simple frequency counts over historical data.

To the best of our knowledge, a foundational contribution in this field is by Teunter [14], who proposed a heuristic approach for the multi-job repair kit problem. In his formulation, each job may require multiple spare parts, potentially in more than one unit. For simulation purposes, $p_{sp}^{t_j}$ is sampled from a uniform distribution $P \sim \mathcal{U}(0, 0.20)$.

Building on this, Bijvank et al. [8] extended the model by assuming that unused spare parts are retained in the repair kit when a job is incomplete and introduced a formal definition of the job fill rate as the proportion of jobs completed without stockouts or revisits. Their greedy algorithm, shown to outperform Teunter's, achieves near-optimal results in settings where exact computation is tractable. In their experiments, $p_{sp}^{t_j}$ is sampled from a uniform distribution scaled by the maximum quantity sp of spare parts required for task t_j .

Practical considerations are introduced in Prak et al. [9], who incorporate fixed ordering costs and lead times into the optimization model. This makes the framework more applicable to real-world contexts where immediate spare part availability in stock is not guaranteed. Their empirical validation uses industrial data from an equipment manufacturer; unfortunately, the method for estimating $p_{sp}^{t_j}$ is not explicitly detailed.

A similar setting is explored in [10], which adds inventory budget constraints and models replenishment frequency. Their two-phase solution combines integer linear programming to optimize the composition of the repair kit with stochastic simulation to test the robustness of the solution under different demand scenarios. The demand distribution is derived from historical data in line with Teunter's methodology.

Recent heuristic-based approaches have also explored spare part list optimization from a different angle. For instance, Zhang et al. [15] propose a modified Pareto Ant Colony Optimization algorithm to generate spare part configurations that balance reliability, cost, and list size through Monte Carlo simulation. Their work focuses on system-level reliability modeling rather than historical task data, exemplifying how bio-inspired heuristics can support spare part planning under uncertainty.

The aforementioned studies address variants of the combinatorial optimization problem under the assumption that no information is

available about the condition of malfunctioning appliances prior to the FST's initial visit to the customer. A more realistic and increasingly relevant scenario involves the availability of Advance Demand Information (ADI)—that is, any data providing insight into the appliance's condition before the technician's first intervention. In this context, Rippe et al. [11] explore the impact of both partial and perfect ADI. In the partial ADI scenario, a subset of spare parts is monitored by sensors; in the perfect ADI case, a failure can be precisely linked to a specific part within that subset.

In their numerical study, these forms of ADI are modeled by assigning a probability $p_{sp}^{ij} = 1$ to any spare part sp for which a failure is reported in advance. For untracked components, p_{sp}^{ij} is sampled from a uniform distribution. The results show that even a simple greedy heuristic—similar to those proposed in [9,14]—can leverage ADI to reduce inventory holding costs without compromising service levels. Moreover, the simulations provide insights into which types of parts benefit most from the availability of ADI.

It is important to note that, although enriched by ADI, the problem addressed in [11] remains a combinatorial optimization problem and differs from earlier formulations only in its inputs. ADI enables the inclusion of additional constraints that guide the selection of spare parts to be included in the repair kit.

In contrast, our setting assumes an imperfect and unstructured form of ADI. Rather than relying on explicit failure codes or sensor outputs, we make use of free-text fault descriptions provided prior to the technician's visit. This unstructured information is then processed using NLP techniques to estimate the probability of spare part usage for a given task.

In a recent follow-up study, Rippe et al. [13] explore a realistic scenario involving partially imperfect ADI. They note that sensors are typically designed to monitor the functioning of assemblies or subsystems rather than individual spare parts. As a result, the fault codes generated by these sensors often indicate issues at the functional level and may correspond to multiple components within the affected assembly. In their numerical experiments, this is modeled by sampling the probability p_{sp}^{ij} from different uniform distributions, depending on whether the spare part sp is part of an assembly associated with a reported fault code or not.

In another extension, Pham et al. [12] address the repair kit problem jointly with FST scheduling and customer routing. They show that incorporating perfect ADI into their model can significantly reduce holding, penalty, and routing costs by improving planning accuracy.

Across all these studies on repair kit problem, one common thread is the reliance on historical spare part usage probability p_{sp}^{ij} . This approach depends heavily on the availability of structured, labeled data being consistently recorded by FSTs. Furthermore, it is not made explicit how this probability estimation is computed in practice. Finally, none of these works make use of unstructured information, such as fault descriptions, or explore learning-based prediction of spare part needs at the task level. This is where our work diverges significantly. Rather than formulating the problem as a combinatorial optimization task with predefined part demand probabilities, we address the problem of predicting the relevant spare parts for a specific repair task. Our system leverages all available structured and unstructured data, including appliance metadata and free-text fault reports, using Deep Learning (DL) and NLP to generate a ranked list of likely needed parts. This approach provides a probabilistic proxy for p_{sp}^{ij} , enabling smarter, data-driven repair kit planning.

While our work is not directly comparable with traditional repair kit optimization methods, it serves as a powerful complementary tool. Compared to the repair kit problem, our approach operates upstream in the decision pipeline and is aimed at estimating in advance which spare parts are likely to be needed to complete a single repair task. The output of our model is a ranked list of candidate spare parts and is intended to feed the repair kit problem solver that may possibly be designed

to incorporate additional constraints, e.g., inventory capacity, maintenance urgency, spare part costs. By improving part prediction accuracy before a technician's first visit, our method reduces over-provisioning, unnecessary revisits, and the need for heuristic estimation of part usage.

Beyond repair kit optimization, related research areas include predictive maintenance, inventory management optimization, spare parts demand forecasting, and additive manufacturing solutions for spare parts logistics. Predictive maintenance aims to anticipate equipment failures before they occur using data-driven techniques [16–18], enabling timely interventions and minimizing downtime. Applications span sectors such as renewable energy [19], food processing [20], petrochemicals, and railway systems [21]. For example, Tracht et al. [22] propose using condition monitoring data to forecast wind turbine failures and pre-position spare parts. In the aerospace domain, Dangut et al. [23] develop a predictive model to identify in advance which aircraft components are likely to require maintenance or replacement. By framing the task as an imbalanced binary classification problem, they combine data resampling and cost-sensitive ensemble learning to improve the detection of rare but critical failures, supporting better maintenance planning and operational safety. However, these methods focus on preventive action, while our approach is designed for the reactive stage, once a fault has already occurred.

Inventory management optimization focuses on enhancing the availability of spare parts while minimizing storage and procurement costs. Recent approaches address this challenge by jointly coordinating inventory policies and maintenance scheduling. Yu et al. [24] proposed a two-stage genetic algorithm that solves a mixed-integer nonlinear model combining inventory control and planned maintenance under uncertain failure rates, showing its scalability on large problem instances. Afifi et al. [25] developed a hybrid memetic algorithm that integrates mathematical programming with local search to jointly optimize preventive maintenance schedules and spare parts inventory for independent machines. Their method minimizes the total cost, including maintenance, holding, and ordering, by solving a mixed-integer linear model through a metaheuristic approach, achieving a favorable trade-off between solution quality and computational efficiency.

Spare parts demand forecasting, on the other hand, aims to estimate future part usage over a given time horizon [2,26–28]. These models optimize inventory levels at a strategic or tactical level, whereas our method operates at the operational level, providing real-time predictions tailored to individual repair tasks. The issues related to inventory management lie beyond the scope of the current work. Moreover, detailed inventory data (e.g., real-time stock levels and logistic constraints) are not available in our case studies. In our setting, the prediction of which spare parts are most likely needed for a specific repair task is made independently of stock availability. As a consequence, the inventory check, in any case, can be performed by the existing enterprise systems before the intervention is scheduled. In cases where one or more recommended parts are not available, the service visit can be rescheduled or the missing components ordered in advance, thereby avoiding unnecessary on-site visits. From this perspective, our tool can actually support and enhance inventory management decisions: by providing early, task-specific predictions of spare part needs, it allows the company to anticipate procurement actions and align stock levels with upcoming maintenance operations.

De Brito et al. [29] proposed an approach to integrate additive manufacturing into spare parts supply chains by identifying suitable components for 3D printing and optimizing the location of additive manufacturing facilities using a p-median model and mixed-integer linear programming. Their method improves part availability and reduces lead times, as demonstrated in a real-world elevator maintenance case study.

In summary, although our work is informed by adjacent research in operations, maintenance, and forecasting, it addresses a unique challenge: predicting the specific spare parts required for a given repair task based on real-time, multi-modal information—particularly unstructured fault descriptions. This problem remains underexplored in

existing literature and represents a novel contribution to both AI-driven maintenance planning and field service optimization.

2.1. Data-driven approaches for spare parts prediction

To the best of our knowledge, the problem of spare parts prediction from historical data and textual fault descriptions has only been addressed in our earlier work presented in [3]. In that work, we introduced CBR-SPP, a CBR method where repair cases are stored in a structured case base and retrieved based on their similarity to new repair requests. The system first filters cases using appliance metadata, and then ranks them based on the textual similarity between the fault descriptions stored in the cases and the description of the fault under consideration. Once relevant cases are identified, an importance score is assigned to each spare part using a weighted frequency mechanism, where parts associated with more similar cases receive higher priority in the recommendation list. While CBR-SPP effectively reduces unnecessary FST visits, it exhibited three key limitations. First, retrieval-based predictions are inherently constrained to previously observed repair cases, making it difficult to generalize to new failure types or underrepresented appliance models. Second, the need to filter by appliance metadata before analyzing textual similarity limits the flexibility of the system, preventing it from leveraging cross-device patterns that may exist in the repair data. Third, as the case base grows, query efficiency degrades, resulting in longer inference times. These challenges underscore the need for a more scalable and adaptable approach.

To address these issues, we propose DL-SPP.sbert, which replaces explicit case retrieval with a deep learning model trained on historical repair data. Instead of retrieving past cases at inference time, DL-SPP.sbert directly models the relationships between appliance metadata, fault descriptions, and spare parts usage using a Multi-Layer Perceptron (MLP) trained for multi-label classification. By leveraging structured categorical attributes alongside semantic textual representations, DL-SPP.sbert eliminates retrieval bottlenecks and improves generalization to previously unseen cases.

Modern NLP techniques have greatly advanced semantic feature extraction, particularly through Transformer-based architectures [30] such as BERT [31], which generate contextualized embeddings capable of capturing long-range dependencies in text. Sentence-BERT (SBERT) [4] builds on BERT by incorporating siamese and triplet networks to produce sentence-level embeddings optimized for semantic similarity. Unlike standard BERT, SBERT applies a pooling operation to create fixed-size sentence vectors, making it well-suited for text comparison tasks. This architecture has been widely adopted in various fields, including medical diagnostics [32], automated assessment [33], and sentiment classification [34], due to its ability to produce high-quality sentence representations without requiring task-specific fine-tuning.

In this work, we leverage SBERT to generate semantic embeddings of textual fault descriptions and appliance names. These embeddings serve as input features in our DL-SPP.sbert model, allowing the deep learning architecture to effectively capture semantic relationships between fault reports and spare parts requirements.

3. Proposed method

The proposed method addresses the challenge of spare parts prediction for field service repair interventions by framing it as a multilabel classification task. We assume that a database containing historical data on repair and maintenance interventions is available. This database generally includes both textual and categorical attributes describing the details of the appliance, the symptoms of the appliance fault, and the spare parts used by the FST during the repair. This setup is both practical and realistic, since companies that provide repair and maintenance

services typically collect this information as part of their standard operations. During the initial call of the appliance owner to the back office, fault symptoms are recorded alongside details identifying the appliance. After the intervention, the FST documents the details of the repair, including the set of spare parts utilized. This combination of pre- and post-intervention data forms a repair intervention record of the database: different companies can describe the intervention using more or less detailed information, but we assume that at least those illustrated in Table 1 are available in a generic repair and maintenance context. Indeed, this information is available in the real-world case studies analyzed in this paper.

The *Appliance Name* identifies the specific model, serving as a fine-grained identifier. It is reasonable to assume that similar appliances tend to have similar names. The *Appliance Type* represents a broader categorization of appliances, less specific than the *Appliance Name*, with multiple distinct appliances potentially falling under the same type. The *Appliance Group* clusters appliances from the same family, enabling the model to capture relationships between different appliance types within the same group. The *Fault Description* provides a concise summary of the appliance fault, as reported by the owner. Following the repair, the FST logs the identifier codes of the spare parts used during the intervention, as shown in the lower section of Table 1. This integration of textual fields, categorical attributes, and spare part codes is the foundation for the training process of the proposed SPP system. By analyzing the relationships between the appliance characteristics, the fault descriptions, and the spare parts used in past interventions, the system is trained to predict the set of spare parts most likely to be useful to solve a specific repair intervention task.

An overview of the architecture of DL-SPP.sbert is shown in Fig. 1.

The textual fields of each repair intervention, namely *Appliance Name* and *Fault Description*, are processed using SBERT [4] to generate high-quality, multilingual, semantic embeddings. These embeddings provide a dense numerical representation of text, capturing contextual meaning and relationships between fault descriptions and appliance types. The SBERT-generated embeddings are 512-dimensional vectors, one for the *Appliance Name* and one for the *Fault Description*. The categorical attributes, namely *Appliance Type* and *Appliance Group*, are encoded using one-hot encoding. SBERT embeddings and one-hot encoded vectors are concatenated, thus forming the final feature representation used by the MLP classifier. This hybrid feature representation allows DL-SPP.sbert to capture both structured categorical relationships and unstructured semantic meaning.

The concatenated feature vector serves as the input to the MLP designed for the multi-label classification task. Its output layer consists of sigmoid-activated units, with a number of neurons equal to the number of unique spare parts in each dataset. The MLP is trained to predict the probability that each spare part is required for a given repair intervention by minimizing the binary cross-entropy loss. The MLP effectively learns to integrate and prioritize information from textual and categorical features through this process. This enables the system to exploit patterns in the data, such as the tendency of similar appliances with comparable fault descriptions to require overlapping sets of spare parts. Further details on hyperparameters configuration of DL-SPP.sbert are provided in Appendix A.

To generate the final recommendations, the system selects the top K spare parts with the highest predicted probabilities, where K is a configurable parameter. This parameter controls the length of the recommended parts list provided to the FST. Adjusting K , the system seeks to balance two competing objectives: avoiding over-provisioning by limiting the inclusion of unnecessary parts and ensuring that all essential parts are included for repair. A higher value of K increases the likelihood of covering all required parts but leads to longer redundant lists.

The SPP system serves as a decision support tool for FSTs. Although it provides a ranked recommendation of spare parts, the final decision

Table 1
Example of repair intervention record with information available before and after the FST performs the repair.

Before-Repair Intervention			
Appliance Name	Appliance Type	Appliance Group	Fault Description
Front Load K500 Series A	76,852	M-KLM-WA	The door seal is defective and the clothes get caught inside and tear off
After-Repair Intervention: Consumed Spare Parts			
Spare Part 1	Spare Part 2	...	Spare Part N
HK7552	HL3247	...	FN3786

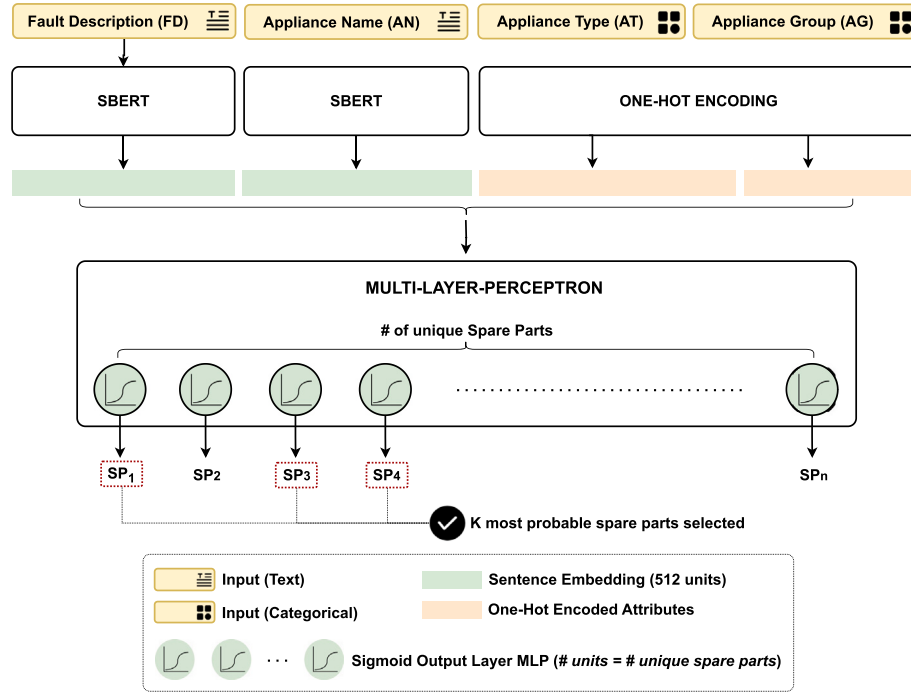


Fig. 1. SPP system architecture diagram.

on which parts to bring to the repair site is left to the discretion of the FSTs, informed by their experience.

4. Case studies

This section presents the experimental analysis conducted on two real-world case studies to validate the proposed DL-SPP.sbert system. The first case study, originally presented in [3], serves as a direct benchmark, and relates to a company specialized in laundry appliances. The second case study, introduced in this manuscript, originates from a Swiss company manufacturing a broader range of household and semi-industrial appliances. Since both datasets come from Swiss companies, the textual records are multilingual, containing entries in Italian, German, and French. This multilingual aspect further underscores the robustness and adaptability of the proposed approach. Although both datasets share the same structure (i.e., feature space), they represent two related but distinct industrial contexts: the types of spare parts and repairable items differ substantially between the two cases. This allows us to evaluate our model in heterogeneous operational settings.

4.1. Datasets description

Both datasets cover the years from 2020 to 2023 and have the same attributes, facilitating direct comparison and analysis. Each record corresponds to a repair intervention on a specific appliance and is divided into pre-intervention and post-intervention information, as described in Section 3. The pre-intervention information, recorded by the company's front-office operators during the initial customer call, includes the fault description (*Fault Description* in Table 1) and appliance details such as the appliance name, type, and group (*Appliance Name*, *Appliance Type*, and *Appliance Group* in Table 1). The post-intervention information, documented by FSTs after the repair, lists the spare parts used during the process. These fields hold particular significance in our case as they are the targets of the SPP system.

To provide a clearer overview of the data used in our experiments, Table 2 presents additional descriptive statistics for both datasets.

The Appliance Metadata section shows that the novel *Dataset B* contains a larger and more heterogeneous set of appliances compared to *Dataset A*, with higher counts of unique names, types, and groups. It

Table 2Main features of *Dataset A* and *Dataset B*, adopted in our experimental analysis.

Descriptive Statistic	Dataset A	Dataset B
Appliance Metadata		
Unique Appliance Names	1200	1800
Unique Appliance Types	350	480
Unique Appliance Groups	45	52
Fault Descriptions (FD)		
Average length (words)	10.3	9.7
Language Distribution (IT / DE / FR)	20 % / 55 % / 25 %	15 % / 65 % / 20 %
Spare Parts		
Number of unique spare parts	2426	4087
Number of repair intervention records by year		
2020	35,432	117,210
2021	35,091	113,402
2022	34,606	107,814
2023	25,871	87,074
Total records	131,000	425,500

also includes a broader range of appliance domains, covering washing, cooking, and refrigeration, while *Dataset A* focuses exclusively on laundry appliances such as washing machines, dryers, and air dryers. This difference reflects the distinct production scope of the two industrial partners and explains the higher level of heterogeneity in *Dataset B*, which makes it particularly suitable for evaluating the generalization ability of the proposed model.

4.2. Data preprocessing

The raw datasets underwent preprocessing to enhance their quality and usability. First, we removed records containing null values in critical fields, such as the *Fault Description*, and discarded duplicate or inconsistent entries to ensure data integrity. Second, textual fields (Appliance Name and Fault Description) were converted to lowercase, stripped of punctuation, and normalized for whitespace, without applying stemming, lemmatization, or stop-word removal, in order to preserve the semantic information required by transformer-based models. Third, spare part descriptions, which were sometimes inconsistent due to multilingual entries (Italian, German, and French) or cosmetic details, were standardized by retaining a single consistent description for each part. Lastly, since the focus of this study is on spare part types rather than their specific identifiers, we consolidated spare parts with different identifiers but the same type. On the other hand, given that the equipment name is known, the specific identifier can be easily retrieved from the type. This process reduced the total number of unique spare part types to 2426 for *Dataset A* and 4087 for *Dataset B*, making the predictive task more manageable while maintaining its relevance for real-world applications.

4.3. Experimental setup

The data sets were divided into training and test sets based on time windows, with repair interventions from 2020 to 2022 used for training and interventions from the first ten months of 2023 used for testing. This temporal split ensures that the model is evaluated on unseen data, closely reflecting real-world deployment scenarios. The main characteristics of the training and test sets are summarized in [Table 3](#), which provides insights into the distribution of tasks grouped by the number of visits required to complete each repair intervention.

We observe that a substantial portion of repair interventions in both datasets required multiple visits, emphasizing the need for an accurate spare parts prediction system. For *Dataset A*, approximately 33 % of the records in both the training and test sets involved two or more visits, while for *Dataset B*, this percentage is 17 %. For the sake of clarity, tasks

Table 3

Descriptions of *Dataset A* and *Dataset B*. Repair intervention tasks are grouped by the number of FST visits. 4+ and 2+ are, respectively, the tasks with a number of visits greater than or equal to 4, and those with a number of visits greater than or equal to 2. For each group of tasks, the mean and the maximum number of spare parts used by the FST to repair the appliance are reported.

# visits	Training Set		Test Set	
	# records	# spare parts Mean Max	# records	# spare parts Mean Max
Dataset A				
1	72,536 (69 %)	1.53 15	17,317 (67 %)	1.54 14
2	25,759 (24 %)	2.77 15	6672 (26 %)	2.81 15
3	5813 (6 %)	3.32 20	1573 (6 %)	3.37 16
4+	1099 (1 %)	4.88 17	309 (1 %)	4.43 14
2+	32,671 (31 %)	2.95 20	8554 (33 %)	2.97 16
Dataset B				
1	282,787 (83 %)	2.65 22	72,994 (84 %)	2.75 15
2	49,021 (15 %)	3.20 19	12,453 (15 %)	3.17 17
3	5316 (2 %)	3.81 19	1328 (2 %)	3.96 12
4+	1374 (1 %)	4.69 20	299 (1 %)	5.09 17
2+	55,711 (17 %)	3.02 22	14,080 (16 %)	3.07 17

that require four or more visits, which represent only a small fraction of the total database (around 1 %), were grouped together. For each dataset, aggregated statistics for tasks requiring at least two visits are provided in the last row of [Table 3](#).

Similar to [\[3\]](#), we exclude single-visit repair interventions from the test sets, as these typically represent cases where FSTs relied on their expertise or used spare parts that are commonly available in their vans. Consequently, this analysis focuses on multiple-visit tasks, where the SPP system can provide the greatest value by reducing the number of additional visits required. For *Dataset A*, the total number of records in the actual test set involving multiple visits is 8554, calculated as the sum of tasks that require 2 (6672), 3 (1573), and 4+ (309) visits. Similarly, for *Dataset B*, the test set consists of 14,080 records, 12,453 requiring two visits, 1328 requiring three visits, and 299 requiring four or more visits.

In the next section, we introduce the evaluation metrics used to quantitatively assess the effectiveness of the SPP system in reducing multiple visits.

4.4. Evaluation metrics

To evaluate the performance of the proposed DL-SPP.sbert system, we use the same metrics proposed in our previous work [\[3\]](#), thus ensuring consistency and comparability. These metrics are designed to quantify the effectiveness of the system in predicting spare parts and its impact on reducing the number of visits required by the FSTs. Below, we briefly describe each metric and provide their mathematical formulations. For further details, we refer readers to [\[3\]](#).

The first metric, Closed Task Percentage (CTP_K), evaluates the proportion of repair tasks successfully completed in a single visit using the predicted list of K spare parts. Formally:

$$CTP_K = \frac{CT_K}{N} \quad (1)$$

where CT_K is the number of closed tasks, and N is the total number of repair tasks in the test set requiring multiple visits. This metric is also defined for tasks requiring i visits as:

$$CTP_K^i = \frac{CT_K^i}{N^i} \quad (2)$$

where CT_K^i and N^i are the number of closed tasks and the total number of tasks requiring i visits, respectively.

The second metric, Saved Visits Percentage ($SV P_K$), measures the proportion of visits saved by the SPP system when a task requiring i visits is completed in a single visit. This metric is computed as:

$$SV P_K = \frac{\sum_{i=2}^I CT_K^i \cdot (i-1)}{\sum_{i=2}^I N^i \cdot (i-1)} \quad (3)$$

where I is the maximum number of visits required to complete a task.

The third metric, Average Recall of Not Closed repair tasks ($recallNC_K$), evaluates the proportion of correctly predicted spare parts for tasks not closed with the spare parts recommended by the SPP system. For a single task t_j , recall is defined as:

$$recallNC_{t_j,K} = \frac{CSP_{t_j,K}}{SP_{t_j}} \quad (4)$$

where $CSP_{t_j,K}$ is the number of correctly predicted spare parts for task t_j using K spare parts, and SP_{t_j} is the total number of spare parts actually used. The average recall for all unresolved tasks (UT) in the test set (TS) is:

$$\overline{recallNC}_K = \frac{\sum_{t_j \in TS} recallNC_{t_j,K}}{UT} \quad (5)$$

The fourth metric, Average Recall ($\overline{recall}_K$), is computed similarly but includes all tasks in the test set, including those closed by exploiting the spare parts list predicted by the SPP system.

The fifth metric, Average Precision ($\overline{precision}_K$), provides insights into the system's allocation efficiency, measuring the proportion of correctly predicted spare parts out of the K suggested. For a single task t_j , precision is defined as:

$$precision_{t_j,K} = \frac{CSP_{t_j,K}}{K} \quad (6)$$

The average precision for all tasks in the test set is:

$$\overline{precision}_K = \frac{\sum_{t_j \in TS} precision_{t_j,K}}{N} \quad (7)$$

These evaluation metrics provide a comprehensive view of the system's performance, with the objective of maximizing each one. In our setting, a cost-benefit trade-off is inherent to the selection of the parameter K , balancing the risk of spare parts overprovisioning against the operational benefits of successful task completion. As K increases, overprovisioning costs tend to rise due to the inclusion of less relevant spare parts, while the probability of successfully completing maintenance tasks also improves. This trade-off is mainly captured by the CTP and $precision$ metrics.

4.5. Comparative approaches for experimental evaluation

Our DL-SPP.sbert is compared with three reference approaches, referred to as CBR-SPP, Freq-SPP, and DL-SPP.bert.

The CBR-SPP system [3] is adopted as the primary baseline for our evaluation, as it represents the current state of the art in spare parts prediction for field service optimization. An overview of the CBR-SPP methodology has been provided in Section 2.

The frequency-based approach (Freq-SPP), widely adopted in the relevant literature [9–13], stems from the insight presented in [14], whereby the probability that a spare part is needed for a task is simply estimated using a frequentist approach on historical data. In our setting, the list of recommended spare parts is generated based on the past cases sharing the same *Appliance Name*. The query granularity is progressively expanded from *Appliance Name* to *Appliance Type* and *Appliance Group* only when the search at a given level fails to retrieve a sufficient number of historical records. Further details on the frequentist approach are

described in [3]. Unlike CBR-SPP, Freq-SPP does not consider the textual information provided with the fault description.

The DL-SPP.bert approach is a variation of the deep learning approach proposed in this manuscript. Compared to the DL-SPP.sbert, the DL-SPP.bert architecture follows a more traditional deep learning pipeline, consistent with early and widely adopted neural approaches to text classification [35]. Since BERT produces contextual embeddings at the token level rather than at the sentence level (as in SBERT), a one-dimensional convolutional neural network (1D-CNN) is applied to extract higher-level local features from the sequence of token embeddings. The resulting feature maps are then flattened and concatenated with one-hot encoded categorical variables, and passed to an MLP for multi-label classification. Implementation details and hyperparameter configuration of DL-SPP.bert are provided in Appendix A.

5. Results

In this section, we analyze the performance of DL-SPP.sbert on the two real-world industrial case studies, comparing it with CBR-SPP, Freq-SPP and DL-SPP.bert. We first present a quantitative evaluation based on the metrics introduced in Section 4.4. We then conduct an ablation study to assess the contribution of different feature groups within the proposed model. Additionally, we provide post-hoc, SHAP-based explanations to illustrate the explainability of the proposed approach in representative maintenance scenarios. Finally, we analyze the impact of concept drift by evaluating model performance over time under different learning strategies.

5.1. Numerical results

Tables 4 and 5 present the performance metrics achieved by the SPP systems for varying numbers of recommended spare parts (K) on datasets A and B, respectively. Each result is averaged over 10 independent training and testing iterations. Standard deviations were consistently low (< 0.5) across all metrics, ensuring stability and reliability; thus, only mean values are reported.

As expected, for all systems, the metrics CTP_K , $SV P_K$, and both $\overline{recall}_K$ and $recallNC_K$ increase as K grows. This trend reflects the ability of all systems to recommend more relevant spare parts as the number of suggestions increases.

Interestingly, the relative ranking among the compared approaches remains largely consistent across both datasets, considering all evaluation metrics and across all values of K . The proposed DL-SPP.sbert demonstrates superior performance across all evaluation settings. The most competitive alternative is DL-SPP.bert, highlighting that neural architectures are essential to attaining satisfactory predictive performance. In general, deep learning-based models outperform the CBR-SPP approach, confirming the advantages of fully data-driven learning strategies. In contrast, the frequency-based model, which does not exploit the textual information provided in the fault descriptions, systematically achieves the lowest performance.

Concerning the values of CTP_K^i for tasks requiring i visits, the proposed DL-SPP.sbert achieves substantial percentages of closed tasks (CTP_{21}^{2+} is around 57 % and 66 % for dataset A and B, respectively). However, as the number of visits increases, CTP_K^i decreases. This trend is likely due to the higher average number of spare parts needed to close tasks involving multiple visits (see Table 3).

All approaches exhibit a tendency toward metric saturation as K increases. The increase in CTP_K and $SV P_K$ is progressively less marked at higher K values, reflecting the decreasing relevance of additional spare parts included in the list. For example, for DL-SPP.sbert, the difference in $SV P_K$ between $K = 18$ and $K = 21$ is merely four and three percentage points on Dataset A and Dataset B, respectively.

The performance gap between DL-SPP.sbert and other approaches is larger on Dataset B than on Dataset A. This suggests that DL-SPP.sbert scales better and is more effective in handling larger and more complex datasets. With Dataset B being four times as large as Dataset A and

Table 4

Values of the evaluation metrics with respect to increasing values of the K recommended spare parts obtained by SPP approaches on *Dataset A* (best values in bold).

K	Model	CTP_K^{2+}	CTP_K^2	CTP_K^3	CTP_K^{4+}	SVP_K	$\overline{recall} \overline{NC}_K$	$\overline{recall}_K$	$\overline{precision}_K$
3	Freq-SPP [14]	11.52	12.33	9.09	6.60	11.01	10.78	23.61	20.32
	CBR-SPP [3]	15.24	16.31	12.02	8.74	14.32	14.02	30.69	26.41
	DL-SPP.bert	18.67	20.24	13.59	10.16	17.36	21.64	36.27	29.20
	DL-SPP.sbert	22.25	22.73	16.84	11.75	20.96	21.88	38.65	30.26
6	Freq-SPP [14]	21.04	22.21	17.68	12.54	20.13	18.36	38.30	17.01
	CBR-SPP [3]	25.01	26.42	20.98	14.89	23.71	21.62	45.14	20.01
	DL-SPP.bert	29.95	31.71	24.35	19.92	28.42	30.39	51.24	21.59
	DL-SPP.sbert	32.97	33.60	27.23	20.92	30.52	32.57	53.09	22.27
9	Freq-SPP [14]	27.92	29.03	24.73	20.24	26.82	22.80	46.50	14.05
	CBR-SPP [3]	32.57	33.86	28.86	23.62	31.36	26.68	54.35	16.43
	DL-SPP.bert	37.17	38.69	32.86	25.61	35.74	35.31	59.36	17.31
	DL-SPP.sbert	40.38	40.95	35.13	27.04	37.89	37.64	61.84	17.97
12	Freq-SPP [14]	34.08	35.23	30.99	24.95	33.02	27.49	54.39	12.54
	CBR-SPP [3]	38.02	39.30	34.58	27.83	36.75	30.61	60.58	13.96
	DL-SPP.bert	43.41	44.93	39.81	28.05	41.84	39.83	65.95	14.71
	DL-SPP.sbert	45.07	46.66	40.72	32.86	42.57	42.11	68.91	15.95
15	Freq-SPP [14]	38.96	40.20	35.91	27.69	37.70	30.25	58.89	11.00
	CBR-SPP [3]	43.24	44.62	39.86	30.74	41.81	33.52	65.27	12.19
	DL-SPP.bert	48.21	49.46	45.71	32.93	46.81	42.93	70.44	12.80
	DL-SPP.sbert	47.71	50.20	45.82	37.35	47.26	45.05	72.37	13.96
18	Freq-SPP [14]	43.53	44.86	40.12	32.35	42.14	32.97	63.07	9.93
	CBR-SPP [3]	47.46	48.91	43.74	35.28	45.98	35.99	68.86	10.84
	DL-SPP.bert	52.73	53.83	50.78	37.80	51.44	45.11	74.05	11.34
	DL-SPP.sbert	53.14	54.50	50.99	40.84	51.72	47.56	76.21	12.49
21	Freq-SPP [14]	48.27	49.50	45.33	36.75	46.79	35.74	67.18	9.14
	CBR-SPP [3]	51.48	52.80	48.32	39.16	50.12	38.28	71.99	9.78
	DL-SPP.bert	56.89	58.01	54.74	42.68	55.61	46.52	76.94	10.19
	DL-SPP.sbert	57.08	58.45	55.07	43.82	55.62	49.33	79.32	11.34

Table 5

Values of the evaluation metrics with respect to increasing values of the K recommended spare parts obtained by SPP approaches on *Dataset B* (best values in bold).

K	Model	CTP_K^{2+}	CTP_K^2	CTP_K^3	CTP_K^{4+}	SVP_K	$\overline{recall} \overline{NC}_K$	$\overline{recall}_K$	$\overline{precision}_K$
3	Freq-SPP [14]	18.43	19.08	13.98	10.73	17.72	13.41	28.98	21.22
	CBR-SPP [3]	22.83	23.75	16.79	11.04	21.81	16.14	34.82	24.87
	DL-SPP.bert	28.35	29.48	21.17	12.46	27.06	22.21	44.26	31.38
	DL-SPP.sbert	32.79	32.90	23.44	16.57	31.55	23.09	47.16	32.79
6	Freq-SPP [14]	28.09	28.92	22.50	17.65	27.17	20.28	42.03	16.57
	CBR-SPP [3]	33.00	33.98	26.66	20.40	31.92	24.12	48.40	18.91
	DL-SPP.bert	39.24	40.57	31.09	19.38	37.71	30.74	57.92	22.63
	DL-SPP.sbert	43.62	44.90	33.12	25.11	43.19	33.79	61.48	23.78
9	Freq-SPP [14]	34.37	35.21	29.06	22.49	33.41	25.95	50.65	14.30
	CBR-SPP [3]	40.19	41.28	33.66	23.75	38.93	29.96	57.17	15.81
	DL-SPP.bert	45.77	47.06	38.05	25.61	44.26	36.63	65.63	18.10
	DL-SPP.sbert	51.52	52.85	39.89	30.53	50.03	39.37	69.24	19.75
12	Freq-SPP [14]	39.59	40.46	33.91	28.03	38.61	30.40	57.05	12.52
	CBR-SPP [3]	46.11	47.31	38.55	29.43	44.75	34.61	63.65	13.80
	DL-SPP.bert	51.05	52.31	43.36	31.83	49.58	40.74	70.99	15.28
	DL-SPP.sbert	55.05	56.35	45.55	36.41	53.58	43.89	75.05	16.75
15	Freq-SPP [14]	44.87	45.81	38.83	31.83	43.81	34.31	62.78	11.33
	CBR-SPP [3]	51.04	52.21	43.45	36.12	49.74	38.11	68.45	12.23
	DL-SPP.bert	55.37	56.74	46.56	36.68	53.83	43.85	74.94	13.25
	DL-SPP.sbert	59.64	60.98	50.03	40.52	58.15	47.85	79.71	14.85
18	Freq-SPP [14]	48.83	49.84	42.42	34.60	47.67	36.94	66.63	10.23
	CBR-SPP [3]	54.91	56.11	47.14	39.46	53.55	40.82	71.97	10.89
	DL-SPP.bert	59.58	60.90	51.17	41.18	58.08	46.12	78.22	11.73
	DL-SPP.sbert	62.47	63.82	53.78	43.83	60.94	50.62	82.65	12.45
21	Freq-SPP [14]	52.81	53.94	45.62	37.37	51.56	39.33	70.12	9.41
	CBR-SPP [3]	56.97	59.98	50.02	42.25	56.15	42.13	73.09	9.45
	DL-SPP.bert	63.08	64.41	55.08	42.56	61.53	47.83	80.74	10.49
	DL-SPP.sbert	65.76	67.69	57.81	47.06	64.22	51.12	84.57	11.25

containing nearly twice the number of distinct spare parts, DL-SPP.sbert demonstrates its ability to handle increased dataset complexity. DL-SPP.sbert achieves up to 55 % visit reduction on *Dataset A* and up to 64 % on *Dataset B* at $K = 21$, demonstrating its strong capability to significantly reduce unnecessary visits.

In terms of precision, DL-SPP.sbert shows a marked improvement over the state of art CBR-SPP approach, particularly for lower values of K . This advantage is critical for minimizing overprovisioning, as fewer irrelevant spare parts are recommended. However, as K increases, the precision gap between the two systems narrows, reflecting the increasing inclusion of less relevant spare parts in the recommendations generated by both systems.

A deeper analysis of the $\overline{precision}_K - \overline{recall}_K$ curves (Fig. 2) provides further insights into models' performance.

Fig. 2 illustrates the expected inverse relationship between precision and recall across all models: increasing K enhances recall but reduces precision as more irrelevant parts are included. The DL-SPP.sbert curve consistently dominates the curve related to the other approaches.

The tendency toward saturation in evaluation metric values, observed earlier, becomes particularly evident in the increasing density of points at higher values of K .

These observations confirm that DL-SPP.sbert consistently retrieves more relevant spare parts while maintaining higher precision, particularly in lower K regimes. However, as K increases, all models tend to include fewer relevant parts, highlighting the inherent trade-off between precision and recall in this task, where each repair typically requires an average of three spare parts.

Beyond numerical metrics, a key distinction between deep learning-based models and alternative baseline approaches lies in their training and inference procedures. Freq-SPP and CBR-SPP, being based on frequency-based and case-based reasoning approaches, respectively, generate recommendations by matching or comparing new queries with past repair cases. These approaches do not require a training phase as predictions are derived directly from historical data. In contrast, DL-SPP models, being neural network-based systems, requires a training phase to learn their parameters. Training DL-SPP.sbert, for example, takes approximately 113.44 ± 2.30 seconds on *Dataset A* and 432.05 ± 3.12

seconds on *Dataset B* using an NVIDIA GeForce RTX 3060 Laptop GPU. Despite this initial overhead, DL-SPP.sbert offers significantly faster inference time, requiring less than 0.1 s per record: the inference time depends only on the architecture of the neural network and not on the size of the dataset. Conversely, the inference time of the state of art approach CBR-SPP is dependent on the size of the case base, taking approximately 0.3 s for *Dataset A* and 0.9 s for *Dataset B*. As the case base grows, CBR-SPP's inference time increases proportionally, making DL-SPP.sbert a more scalable and efficient solution for real-time predictions and large-scale datasets.

While the superiority of deep learning-based approaches over the other methods is evident, we further investigated whether the performance difference between DL-SPP.sbert and DL-SPP.bert is statistically significant. The statistical significance analysis, reported in Appendix B, confirms that DL-SPP.sbert significantly outperforms DL-SPP.bert in most settings. Therefore, in the following, we focus our discussion on the proposed DL-SPP.sbert.

5.2. Ablation study

To better understand the contribution of each feature to the performance of the DL-SPP.sbert system, we conducted an ablation study by systematically excluding different features and analyzing their impact on the Saved Visits Percentage ($SV P_K$). The results on *Dataset A* and *Dataset B* are reported in Figs. 3 and 4, respectively.

For *Dataset A*, the best results in terms of $SV P_K$ are achieved when all the features (i.e., Fault Description FD, Appliance Name AN, Appliance Type AT and Appliance Group AG) are included in the model, particularly for values of K up to 12. Removing individual features allows us to assess their relative importance.

Focusing on configurations involving three features in Fig. 3, we observe that the removal of FD (i.e., configuration -- AN AT AG, where -- indicates that the corresponding feature is not included in the set) leads to a sharp decrease in performance, with a drop of approximately 10 % $SV P_K$ across all values of K . This highlights that FD is a crucial feature, as it is the only one that provides information about the nature of the appliance malfunction. Among the configurations involving three features,

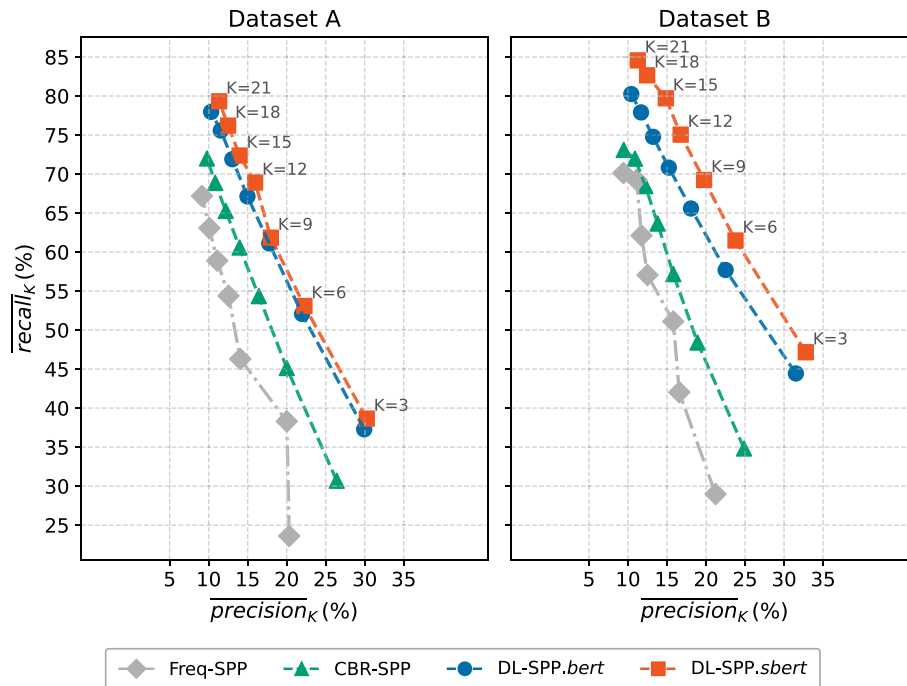


Fig. 2. $\overline{precision}_K - \overline{recall}_K$ curves for Freq-SPP, CBR-SPP, DL-SPP.bert and DL-SPP.sbert on *Dataset A* and *Dataset B*. Each point corresponds to a different K , showing the trade-off between $\overline{precision}_K$ and $\overline{recall}_K$. To improve figure readability, the values of K are annotated only for one curve. All curves follow the same progression.

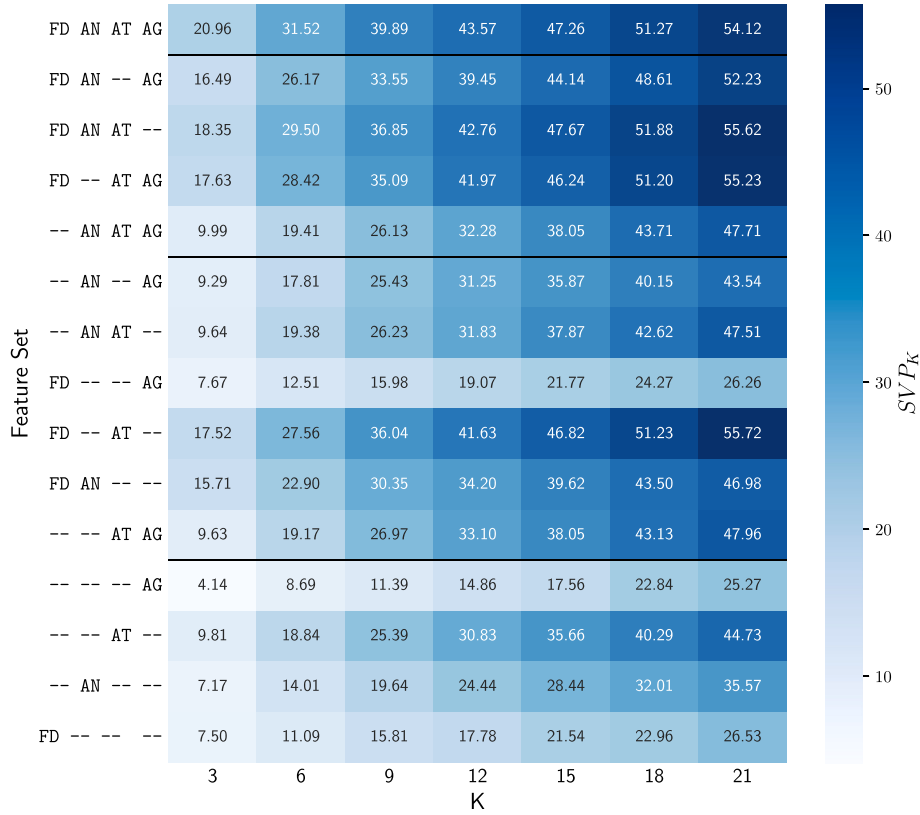


Fig. 3. Ablation study results. $SV P_K$ as a function of the set of features used to train the MLP on Dataset A. *FD* = Fault Description; *AN* = Appliance Name; *AT* = Appliance Type; *AG* = Appliance Group. -- indicates that the corresponding feature is not included in that set.

FD AN AT -- achieves the best results, indicating that excluding AG results in a performance drop of only about 2 % $SV P_K$ for K between 3 and 12, while for larger values of K , the difference becomes negligible. Analyzing the three rows that contain FD in combination with two of the three appliance-related features (AN, AT, AG), we find that AT is the most informative, followed by AN. This is likely due to the fact that AT provides an intermediate level of granularity, more specific than AG but more general than AN, making it a more effective predictor.

These results suggest that FD and AT should be prioritized due to their strong contribution to predictive accuracy. In contrast, AG could be omitted in scenarios with computational constraints, as its contribution is limited.

Similar trends are observed in Dataset B, where the best-performing configuration is once again the one that includes all four features (FD, AN, AT, AG).

Analyzing the configurations involving three features, the removal of FD (-- AN AT AG) once again leads to a sharp drop of approximately 10 % $SV P_K$, confirming its critical role. However, unlike in Dataset A, the gap between the three features characterizing the appliance is narrower. The exclusion of AG (FD AN AT --) has the least impact, causing a slight decrease in $SV P_K$ of approximately 3 % for K between 3 and 12, and becoming almost negligible (1 %) for $K \geq 15$. This suggests that AG adds value, but is not a decisive factor for predictive accuracy.

Observing the configurations involving a single feature, AG is again the least contributive. This is further confirmed by the analysis of the configurations with two features, where the best-performing pairs are FD AN --, followed closely by FD -- AT --. This aligns with the observation that AN is the most informative feature overall, slightly outperforming AT when considered individually. The worst-performing configuration is, once again, FD -- -- AG, reinforcing the idea that AG alone is not a strong predictor.

From these results, we conclude that FD and AN should be prioritized in the model's design, as they have the strongest impact on prediction accuracy. Furthermore, AG could be removed in computationally constrained environments with an acceptable trade-off (1-3 % $SV P_K$ reduction).

The ablation study highlights the relative importance of each characteristic in the prediction process. FD consistently emerges as the most critical feature, as its removal results in a significant drop in predictive accuracy. Among the appliance-related attributes, AT proves to be the most valuable feature in Dataset A, while AN has a slightly stronger impact in Dataset B. The AG feature provides only marginal improvements and could be omitted when computational cost is a concern.

5.3. Local explainability analysis

The ablation study reveals the contribution of each feature to the overall performance of DL-SPP.sbert from a global perspective. In this section, we provide examples of post-hoc, SHAP-based explanations [6] for individual representative maintenance scenarios. This local explainability analysis aims to provide an intuitive understanding of DL-SPP.sbert's behavior at the level of individual maintenance tasks. For each prediction, we compute the Shapley values over the four input blocks, namely AG, AT, AN, and FD. By comparing the relative importance of these inputs, we can highlight which factors play a dominant role in shaping the model's output in each case.

We believe that, among the four feature blocks, FD is particularly interesting, since it represents unstructured input provided by human operators. Its content and quality may vary depending on how faults are described, and can potentially be improved through specific guidelines or company policies aimed at standardizing maintenance reporting. Specifically, we analyzed two representative examples of repair

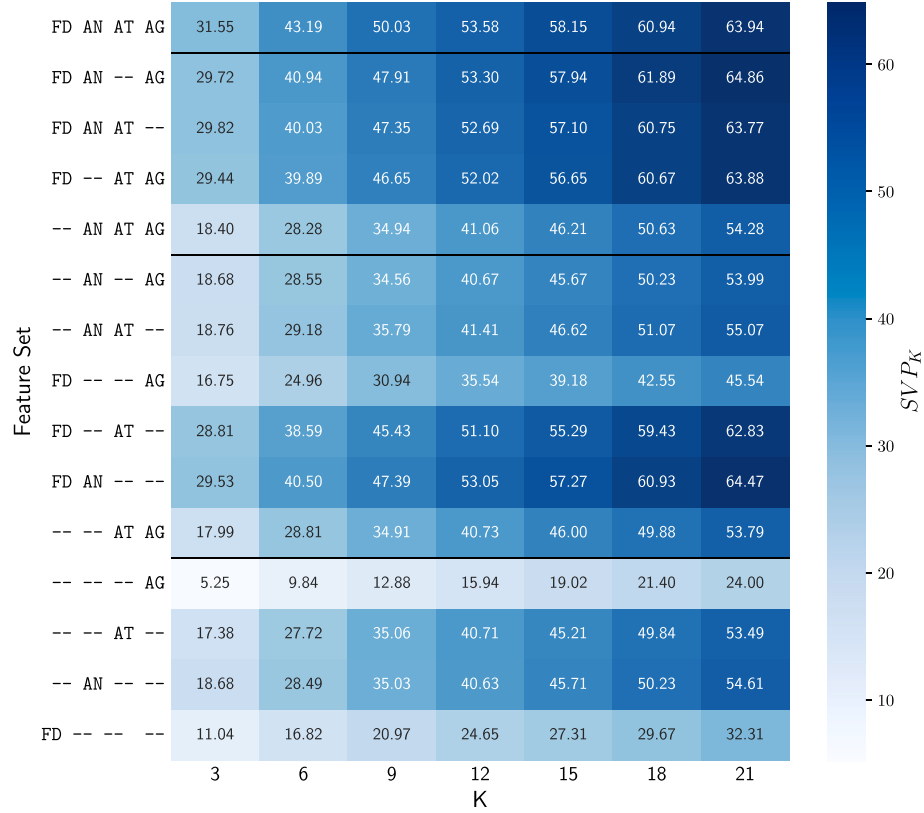


Fig. 4. Ablation study results. $SV P_K$ as a function of the set of features used to train the MLP on Dataset B. *FD* = Fault Description; *AN* = Appliance Name; *AT* = Appliance Type; *AG* = Appliance Group. -- indicates that the corresponding feature is not included in that set.

Table 6

Representative instances of repair interventions with vague (*RI-1*) and informative (*RI-2*) fault descriptions.

RI	AG	AN	AT	FD
RI-1				
Vague FD	Geschirrspüler (Dishwasher)	Geschirrspüler adora spülen v2000 vollintegrierbar home gleitblech lcd display (<i>Adora dishwasher V2000, fully integrated, home, sliding panel, LCD display</i>)	41,093	Der geschirrspüler pfeift dauernd nach letztem servicegang (<i>the dishwasher keeps whistling after the last service cycle</i>)
RI-2				
Informative FD	Geschirrspüler (Dishwasher)	Geschirrspüler adora 55 (<i>Dishwasher Adora 55</i>)	182	Rollen von oberer geschirrkorb fällt aus korbschiene, besteckkorb ersetzen, filtersieb gebrochen (<i>wheels of the upper basket fall out of the rail, cutlery basket to be replaced, filter broken</i>)

interventions that illustrate complementary scenarios: a repair with vague fault description (*RI-1*) and a repair with informative fault description (*RI-2*). Table 6 reports the values of the input features for the two cases. English translation (within parentheses, in italic) is included for reader convenience and is not part of the original report.

Our DL-SPP.sbert is leveraged to provide the most probable spare parts for each task: each suggested spare part is characterized by a name and an associated probability value, which determines its ranking in the recommendation list. In addition, we have access to the ground truth information indicating whether each part was actually used during the intervention. For each predicted spare part, we also compute the Shapley values corresponding to the four input blocks, namely ϕ_{AG} , ϕ_{AT} , ϕ_{AN} , and ϕ_{FD} , which represent the additive importance scores of each feature.

Tables 7 and 8 report the results of the local explainability analysis for the two representative cases. For the sake of clarity, each table reports only the ground-truth parts, along with their actual rank. The output probability value of a part (referred to as Part Prob) is obtained as the sum of these Shapley values and the base value, computed as

the average predicted probability over the examples in the background dataset (i.e., the training set).

In the case of *RI-1* (Table 7), the model recommends three out of five ground-truth parts within the top six predictions (the remaining two are ranked #11 and #153). Conversely, in the case of *RI-2* (Table 8) the model recommends all the necessary spare parts within the top six predictions.

Our local explainability analysis reveals an interesting insight in terms of error analysis. In the case where *FD* is vague (*RI-1*), the distribution of Shapley values indicates that none of the input features exerts a clearly dominant influence on the model's predictions. Moreover, a low probability value is assigned even to the top-ranked parts, suggesting that the model exhibits limited confidence in its output. Although the system is unable to automatically close the task in such a case (one required spare part would not be suggested, ranking #153), it still manages to compensate for the lack of clarity in the *FD* by leveraging the remaining features to predict other relevant spare parts.

Table 7

Prediction of DL-SPP.*sbert* for ground truth spare parts for *RI-1* (vague FD), along with Shapley values of input attributes.

Part Name	Part Prob (Ranking)	Base value	AG	AT	AN	FD
steuerteil (<i>control module</i>)	0.108 (2)	0.014	0.020	0.025	0.044	0.009
leistungsteil adora (<i>power module Adora</i>)	0.108 (3)	0.003	0.027	0.028	0.060	−0.007
kabelbaum türe türöffner (<i>door wiring harness / door opener cable set</i>)	0.056 (6)	0.001	0.010	0.010	0.026	0.011
lcd bildschirm (<i>LCD display</i>)	0.032 (11)	0.002	0.010	0.008	0.025	−0.010
sieb (<i>filter, generic</i>)	0.001 (153)	0.001	0.002	0.001	0.002	−0.004

Table 8

Prediction of DL-SPP.*sbert* for ground truth spare parts for *RI-2* (informative FD), along with Shapley values of input attributes.

Part Name	Part Prob (Ranking)	Base value	AG	AT	AN	FD
vorm rollenhalter (<i>front roller holder</i>)	0.686 (1)	0.008	0.053	0.066	0.038	0.522
grobsieb (<i>coarse filter</i>)	0.416 (2)	0.002	0.047	0.002	0.074	0.294
hydrophil besteck-korb (<i>hydrophilic cutlery basket</i>)	0.347 (4)	0.001	0.033	0.040	0.052	0.223

In contrast, in the *RI-2* case, where the fault description is highly informative, the importance attributed to FD, as evidenced by the Shapley values, dominates the prediction process. The tokens contained in the FD feature, such as wheels, rail, cutlery basket, and filter, correspond directly to the retrieved components, demonstrating a strong alignment between the semantic content of the input text and the parts suggested by the model. The relatively high probability values assigned to the top-ranked parts further support this interpretation, indicating that the model exploits textual patterns that, in this case, exhibit a clear relationship with the predicted spare parts.

5.4. Concept drift analysis

In the considered practical maintenance setting, new fault types and new equipment models emerge continuously. More generally, classification tasks in evolving settings are typically affected by the phenomenon of concept drift [5] which can be broadly defined as a change in the underlying data generating process. Due to concept drift, the performance of a classifier, trained solely on an initial set of data, may deteriorate over time. To counteract this effect and preserve or improve classification accuracy, it is essential to adopt appropriate incremental learning strategies.

To assess the impact of concept drift on our SPP system, we devise a new experimental setup as follows: we split each dataset into 24 consecutive, non-overlapping chunks of two months each. We then merged the first 8 chunks (i.e., covering the first 16 months of data) into a single initial training set. The remaining 16 chunks are used for testing the spare parts prediction system's performance over time, and possibly for updating the system itself. For this experimental evaluation, *Dataset A* comprises 43,693 instances in the initial training set, and it has an average chunk size of 5462 samples; *Dataset B* comprises 141,170 training instances and it has an average chunk size of 17,647 samples. Importantly, at step n the classification system is evaluated on chunk n , but can be trained only on data up to chunk $n - 1$, thus avoiding any form of temporal leakage.

We compared the performance of four different learning strategies, described in the following.

- **Baseline (BL):** the spare parts prediction system is trained just once, on the initial training set.

- **Retrain from scratch (RFS):** at each available chunk of training data, the training set is extended with the new chunk and the spare parts prediction system is re-trained from scratch. In data stream mining this strategy is referred to as prequential evaluation or interleaved-test-then-train.
- **First-In-First-Out (FIFO):** at each available chunk of training data the model is retrained from scratch in a sliding window fashion; specifically, the training set is updated by adding the new chunk and removing the oldest chunk of data.
- **Fine-Tuning (FT):** at each available chunk of training data, the spare parts prediction system is updated. In practice, the deep learning model trained on past data is further fine-tuned using the most recent chunk.

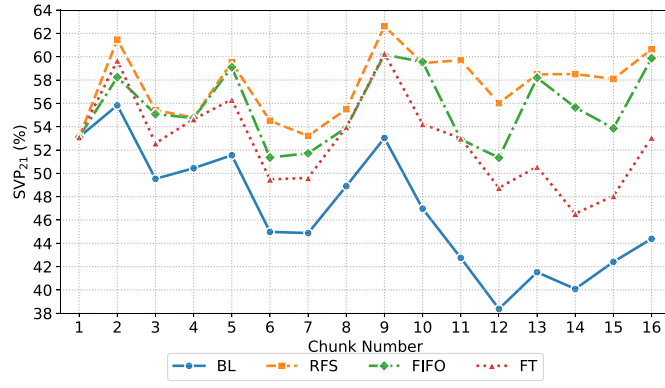
The four strategies are evaluated using SVP as a metric to assess the effectiveness of the SPP system. For this experiment, we fixed the recommendation list length to $K = 21$ ($SV P_{21}$). As discussed in Section 5.1, DL-SPP.*sbert* exhibits a tendency toward metric saturation as K increases. Choosing a relatively high value of K allows us to better isolate the effect of concept drift, since performance variations can be attributed to the update strategy rather than to the truncation of the recommendation list. Moreover, from an operational perspective, this remains a reasonable and valid choice, as it offers a comprehensive set of potentially useful spare parts to the FSE while limiting unnecessary over-provisioning.

Fig. 5 reports the trends in the performance metric over time across the two datasets.

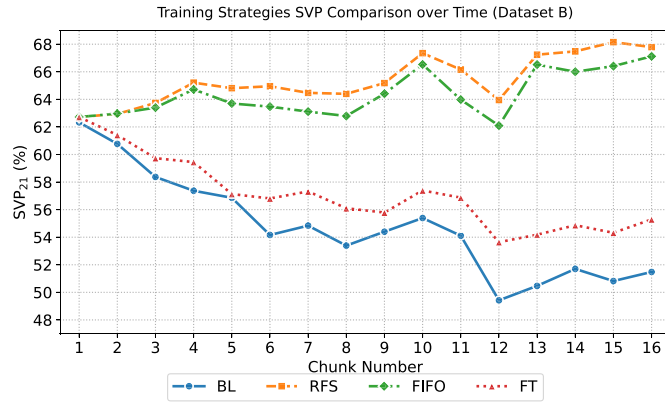
On both datasets, the baseline approach (BL) clearly shows a degradation in SVP values over time. The system is never updated and thus fails to model the evolving data distribution. The different degradation patterns observed for the BL strategy on the two datasets suggest that they may be affected by different types or intensities of concept drift. A more detailed characterization of the underlying drift phenomena therefore represents an interesting direction for future research.

RFS learning strategy provides consistently higher accuracy compared to BL. By retraining the model from scratch on all historical data whenever a new chunk is available, it leverages both past knowledge and the most recent information, thereby maintaining better alignment with the evolving data distribution. It is worth underlining that SVP values still exhibit some variability across chunks. For *Dataset A*, the trend is not strictly monotonic, yet the values remain consistently comparable to or higher than the initial level (SVP around 53 %), confirming the overall robustness and usefulness of the approach. Conversely, on *Dataset B*, SVP shows a more stable and gradually increasing pattern, suggesting that a larger amount of training data supports better adaptation over time. A possible motivation for the lower variance observed for *Dataset B* compared to *Dataset A* may lie in its greater data volume.

The learning strategy based on sliding window (FIFO) yields interesting results: on *Dataset A*, SVP values generally trace those of RFS, confirming the method's ability to preserve good predictive accuracy while limiting the training set size. On *Dataset B*, SVP values are slightly below those obtained with RFS, indicating that the larger amount of



(a) Dataset A



(b) Dataset B

Fig. 5. Comparison of learning strategies along the monitoring campaign in terms of SVP_{21} .

available data mitigates the information loss caused by the windowing mechanism.

FT learning strategy only partially mitigates concept drift: it generally outperforms BL on both datasets, but fails to keep pace with the RFS, which entails periodic full re-training.

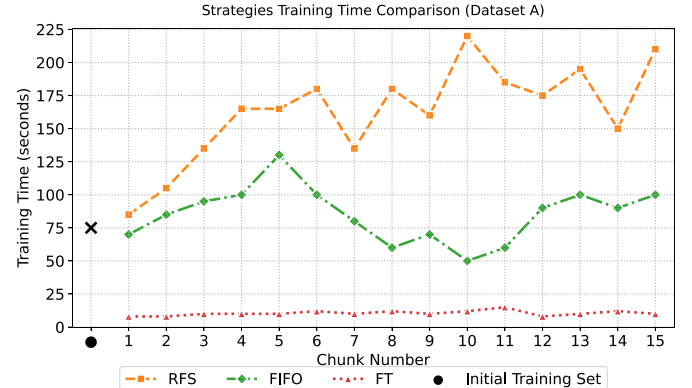
Fig. 6 reports the training time measured at each chunk for the four learning strategies on the two datasets. The BL strategy is not reported as it does not involve any form of adaptation.

The results highlight the expected trade-offs: RFS requires the highest training times, and they tend to increase over time as the training set grows. FIFO, instead, shows nearly constant training times across chunks, yet it still entails retraining the model from scratch at each update. FT achieves adaptation with much lower computational overhead, making it the most efficient strategy in terms of computational cost.

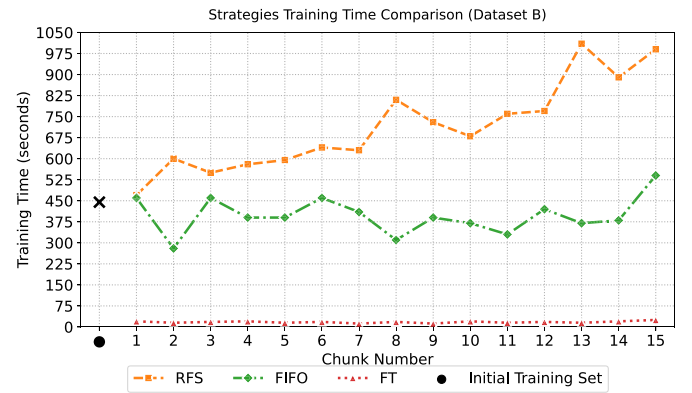
In summary, the concept drift analysis confirms the limitations of static models and the need for adaptive learning in dynamic maintenance scenarios. Among the strategies, retraining from scratch generally delivers the best overall performance and is therefore recommended in practice. Although it is the most computationally intensive option, the cost remains definitely acceptable in our case studies since training times are only on the order of minutes.

6. Conclusion

In one of our previous works, we introduced CBR-SPP (Case-Based Reasoning Spare Parts Prediction) [3], a system that recommends spare parts by retrieving past repair cases from a structured case base based on their similarity to the input repair request. While CBR-SPP demonstrated strong predictive performance and effectively reduced unnecessary revisits by Field Service Technicians (FSTs), its reliance on structured



(a) Dataset A



(b) Dataset B

Fig. 6. Comparison of learning strategies along the monitoring campaign in terms of training time.

retrieval introduces computational overhead and limits its generalizability to novel repair scenarios. As the case base expands, retrieval times increase proportionally, making real-time inference increasingly challenging.

To address these limitations, in this paper we have proposed DL-SPP.sbert, a deep learning-based approach to spare parts prediction aimed at supporting FSTs by reducing the number of multiple-visit repair interventions. DL-SPP.sbert eliminates the need for explicit case retrieval by employing a Multi-Layer Perceptron trained directly on historical repair data. In contrast to CBR-SPP, which retrieves similar cases at inference time, DL-SPP.sbert learns direct relationships among appliance metadata, fault descriptions, and spare parts usage, thus enabling faster, more scalable, and more generalizable predictions.

In an extensive experimental evaluation, DL-SPP.sbert is compared against three representative baselines: Freq-SPP, DL-SPP.bert, and the above mentioned CBR-SPP. Freq-SPP represents a simple yet widely adopted baseline that estimates spare part relevance through frequency statistics computed over historical repair data. DL-SPP.bert follows a traditional deep learning pipeline for text-based prediction, combining token-level contextual embeddings extracted by BERT with convolutional feature extraction and structured appliance metadata to perform multi-label spare parts prediction.

To evaluate our SPP system, we analyzed two real-world case studies using historical repair data from appliance service companies that provide maintenance and repair services to customers. The first case study was based on the same dataset used in the CBR-SPP evaluation, enabling a direct performance comparison between the two systems. The second case study employed data from a different company within the same

industry, allowing us to assess the generalizability of both approaches across diverse operational contexts.

Experimental results highlight the significant advantages of DL-SPP.sbert over other approaches and, in particular, over the state of art CBR-SPP solution. When suggesting a list of $K = 9$ spare parts, DL-SPP.sbert achieves a notable improvement in Closed Task Percentage (CTP_K), with gains of +10.1 % in the first case study and +11.3 % in the second one, compared to CBR-SPP. These results confirm DL-SPP.sbert's superior ability to recommend the correct spare parts. Notably, when recommending $K = 21$ spare parts, DL-SPP.sbert reduces multiple-visit repair interventions by up to 64 % thus demonstrating its effectiveness in optimizing field service operations.

A deeper analysis of the $\overline{precision}_K - \overline{recall}_K$ trade-off confirms that DL-SPP.sbert consistently outperforms baseline approaches across all values of K , retrieving a larger proportion of relevant spare parts while minimizing over-provisioning.

In addition to the overall performance comparison, we conducted a detailed analysis of factors that are crucial for the adoption of the proposed system in industrial contexts, namely model explainability and robustness to concept drift in evolving operational settings.

First, we conducted an ablation study to assess the contribution of individual features to DL-SPP.sbert's predictive accuracy. Among the structured appliance attributes, the study reveals that *Appliance Type* and *Appliance Name* are the most informative features. In the first case study, *Appliance Type* emerges as the strongest feature, while in the second case study, *Appliance Name* shows a slightly greater impact. In contrast, *Appliance Group* provides only marginal improvements, and its exclusion results in a minimal decrease in predictive performance, particularly for larger values of K . This suggests that, in computationally constrained environments, *Appliance Group* could be omitted with an acceptable trade-off of a 13 %–3 % reduction in saved visits percentage. Most importantly, the study confirms that fault descriptions play a fundamental role, as removing them leads to a sharp decline in performance across both case studies, with a drop of approximately 10 % in saved visits percentage. This highlights the necessity of textual fault descriptions in capturing essential repair-related information.

These observations are consistent with the local explainability analysis presented in this work, where SHAP-based local explanations reveal the critical role of fault description quality in shaping model behavior. We considered representative examples of repair intervention: on the one hand, when the fault description is vague, the contribution of individual features is more evenly distributed and no dominant driver emerges, resulting in lower confidence scores and partial coverage of the required spare parts. On the other hand, when the fault description is informative, SHAP values show a clear dominance of the textual feature, with semantically meaningful tokens directly driving the selection of the correct spare parts. This results in higher prediction confidence and complete coverage of the required components within the top-ranked recommendations.

Second, we analyze the impact of concept drift in evolving maintenance settings. The experimental results from both case studies confirm that static models rapidly degrade when exposed to changing data distributions, highlighting the necessity of adaptive learning strategies. The analysis reveals that periodically retraining the model by incorporating the most recently collected data is crucial to maintaining, or even improving, the performance of the SPP system over time. While this introduces additional computational costs compared to a static system, such costs remain acceptable for real-world industrial deployments, including on-premises environments.

Overall, DL-SPP.sbert represents a significant advancement in spare parts prediction, surpassing comparative approaches in accuracy, efficiency, and scalability. By reducing unnecessary technician revisits, improving spare parts provisioning, and enabling real-time inference, DL-SPP.sbert offers a robust, AI-driven solution for modern field service optimization. Its adoption in real-world repair and maintenance operations can lead to cost savings, improved customer satisfaction,

and enhanced operational efficiency, reinforcing the impact of AI-driven decision support systems in industrial applications.

CRedit authorship contribution statement

Alessio Schiavo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Alessandro Renda:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Pietro Ducange:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Francesco Marcelloni:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Implementation details and hyperparameter configuration

In this section, we provide implementation details regarding DL-SPP.sbert and DL-SPP.bert. Implementation details regarding CBR-SPP and Freq-SPP can be found in [3].

Concerning DL-SPP.sbert, the sentence embeddings used in our approach were obtained with a publicly available pre-trained multilingual model, namely `distiluse-base-multilingual-cased-v1`.¹ This model is a distilled version of the multilingual Universal Sentence Encoder, supporting 15 languages, including those present in our datasets (Italian, German, and French). No additional fine-tuning was performed; the model was used in its original configuration through the `SentenceTransformers` Python library (v2.2.2). The embeddings were extracted using default settings (mean pooling over the last transformer layer and 512-dimensional sentence vectors). Each input record was processed independently, and embeddings were computed both for the *Appliance Name* and the *Fault Description* fields. This setup ensures full reproducibility, as the model weights, tokenizer, and encoding pipeline are all publicly available and version-controlled within the Hugging Face model card. After concatenation with one-hot encoded features, the resulting feature vectors have size 1419 for *Dataset A* and 1556 for *Dataset B*.

To determine the optimal architecture of the MLP used for classifying spare parts for each repair intervention, we conducted a comprehensive hyperparameter optimization process. For both datasets *A* and *B*, the last 20 % of the training set was reserved as a validation set to avoid any look-ahead bias during optimization. We used the *Keras Hyperband Tuner* algorithm² to explore configurations such as the number and size of hidden layers, dropout rates, and learning rates. The search ranges and final optimal configurations are summarized in Table A.9.

¹ <https://huggingface.co/sentence-transformers>, visited Dec. 2025

² https://keras.io/keras_tuner/api/tuners/hyperband/, visited Dec. 2025

Table A.9

MLP architecture hyperparameters optimization. Hyperparameter ranges and optimal configurations for Datasets A and B.

Hyperparameter	Range	Optimal Value	
		Dataset A	Dataset B
1st Hidden Layer Units	2048–4096 (step: 512)	4096	4096
2nd Hidden Layer Units	1024–4096 (step: 256)	2048	2048
2nd Hidden Layer Dropout Rate	0.2–0.5 (step: 0.1)	0.4	0.5
Include 3rd Hidden Layer	{True, False}	False	False
3rd Hidden Layer Units	1024–2048 (step: 256)	–	–
3rd Hidden Layer Dropout Rate	0.2–0.5 (step: 0.1)	–	–
Learning Rate	{1e-4, 1e-3, 1e-2}	1e-3	1e-3

Once the optimal architecture was identified, the model was re-trained using the entire training set, including the validation data, to fully leverage the available training data.

For both datasets, the final architecture consists of a first hidden layer with 4096 units, followed by a second hidden layer with 2048 neurons, and an output layer corresponding to the number of unique spare parts in the dataset: 2426 for *Dataset A* and 4087 for *Dataset B*. The models use Rectified Linear Unit (ReLU) activation function in the hidden layers and Sigmoid activation in the output layer to effectively handle the multi-label classification task.

The DL-SPP.sbert model was trained using the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1e-7$) with a learning rate of 0.001 and binary cross-entropy loss. Training was conducted with a batch size of 256 for up to 25 epochs. The batch size was fixed rather than included in the hyperparameter search, as 256 provided a stable trade-off between convergence speed and generalization. Adaptive optimizers such as Adam are known to be relatively insensitive to moderate batch-size variations. During the first phase, 20 % of the training data was held out for validation, and both EarlyStopping (*patience* = 6, *restore_best_weights* = *True*) and *ReduceLROnPlateau* (*factor* = 0.1, *patience* = 3, *min_lr* = $1e-6$) were used to monitor the validation loss. The optimal configuration was reached at epoch 25, after which a final retraining was performed on the entire training set (train + validation) for 25 epochs, maintaining the

same batch size and optimizer settings. In this phase, the learning rate scheduler was retained with *monitor* = ‘loss’ to ensure smoother convergence while preventing gradient plateaus. Weights were initialized with the Glorot uniform initializer (default in Keras), and random seeds were fixed to guarantee reproducibility. All experiments were implemented in TensorFlow 2.10/Keras and executed on an NVIDIA RTX 3060 Laptop GPU (6 GB VRAM).

The DL-SPP.bert variant employs BERT as a frozen contextual encoder. The resulting token embeddings are processed by a one-dimensional convolutional neural network (1D-CNN) to extract higher-level textual features, which are then concatenated with the categorical metadata and fed into the MLP classification head. In detail, for each record, the input text provided to BERT consists of the concatenation of the Appliance Name (AN) and the Fault Description (FD), formatted as “NAME: [AN] ERROR: [FD]”. This unified textual input ensures that the same information used in DL-SPP.sbert is processed in a token-level fashion. The inserted markers help the model distinguish the two text segments, and BERT encodes the entire sequence into a contextual embedding matrix of size $(nT \times d)$, where nT denotes the number of tokens generated by the BERT tokenizer from the input string, and d is the dimensionality of the token embeddings produced by the BERT model. Each row of the matrix therefore corresponds to the contextual embedding of a single token within the combined input describing the appliance name and the fault description.

As for the CNN block, we followed the common practice in neural text classification [35]: a two-layer Conv1D + ReLU operations were used with multiple kernel sizes to capture features of varying granularity, followed by max pooling to aggregate the most informative activations across the sequence. The resulting feature maps are flattened and concatenated with the one-hot encodings of the AT and AG features, as in DL-SPP.sbert. These combined features are then provided to an MLP classifier whose configuration was optimized jointly with the CNN through a hyperparameter search.

Appendix B. Statistical significance analysis

For each dataset, we performed 10 independent training and testing runs of deep learning-based approaches using different random seeds, obtaining a distribution of results for each evaluation metric and each value of K . We carried out a statistical significance analysis based on the main evaluation metric, $SV P_K$, considering values of $K \in \{3, 6, 9, 12, 15, 18, 21\}$. The Freq-SPP baseline was not involved in this

Table B.10

Values of the $SV P$ evaluation metric for CBR-SPP, DL-SPP.bert and DL-SPP.sbert for different values of K , along with their confidence intervals (95 %) where applicable. The p -values associated with the one-sample t -test (CBR-SPP vs DL-SPP.sbert) and two-sample t -test (DL-SPP.bert vs DL-SPP.sbert) are also reported. p -values lower than the significance level $\alpha = 0.05$ are highlighted in bold.

K	CBR-SPP	DL-SPP.bert [CI]	DL-SPP.sbert [CI]	p -value CBR-SPP vs DL-SPP.sbert	p -value DL-SPP.bert vs DL-SPP.sbert
Dataset A					
3	14.32	17.67 [17.38, 17.96]	20.96 [20.65, 21.27]	0.00000	0.00000
6	23.71	28.68 [28.32, 29.05]	30.52 [29.98, 31.46]	0.00000	0.00001
9	31.36	35.93 [35.55, 36.31]	37.89 [37.40, 38.38]	0.00000	0.00550
12	36.75	41.08 [40.62, 41.55]	42.57 [41.94, 43.20]	0.00000	0.00234
15	41.81	46.39 [45.88, 46.90]	47.26 [46.47, 48.05]	0.00000	0.04842
18	45.98	51.76 [51.35, 52.17]	51.72 [50.92, 52.52]	0.00000	0.72437
21	50.12	55.50 [55.19, 55.82]	55.62 [54.76, 56.48]	0.00000	0.59970
Dataset B					
3	21.81	26.63 [26.39, 26.86]	31.55 [31.35, 31.75]	0.00000	0.00000
6	31.92	36.72 [36.49, 36.94]	43.19 [42.93, 43.45]	0.00000	0.00000
9	38.93	43.16 [42.84, 43.48]	50.03 [49.72, 50.34]	0.00000	0.00000
12	44.75	48.51 [48.13, 48.89]	53.58 [53.24, 53.92]	0.00000	0.00000
15	49.74	53.07 [52.75, 53.38]	58.15 [57.79, 58.51]	0.00000	0.00000
18	53.55	57.02 [56.66, 57.38]	60.94 [60.60, 61.28]	0.00000	0.00000
21	56.15	60.36 [60.02, 60.70]	64.22 [63.87, 64.57]	0.00000	0.00000

analysis because it serves as a simple heuristic benchmark rather than a competing learning-based method, and its results are clearly lower in comparison with the other approaches. First, we computed the 95 % confidence interval (CI) for DL-SPP.bert and DL-SPP.sbert. Then, we performed a statistical *t*-test to assess whether the differences between the results of the three spare parts prediction models were statistically significant. We used DL-SPP.sbert as a control model and we statistically compared the results achieved by this model with the ones achieved by the remaining models.

We verified the normality assumption of the *SV**P* distributions obtained by the deep learning-based models across the 10 independent training runs using the Shapiro-Wilk test. The test was conducted separately for all values of *K* on both datasets. For DL-SPP.sbert, *p*-values ranged between 0.35 and 0.85 on *Dataset A* and between 0.09 and 0.93 on *Dataset B*, consistently exceeding the 0.05 significance threshold. Similarly, for DL-SPP.bert, *p*-values ranged between 0.27 and 0.63 on *Dataset A* and between 0.49 and 0.99 on *Dataset B*. These results confirm that the normality assumption holds for both models under all tested conditions, thereby validating the use of Student's *t*-test and supporting the statistical soundness of the reported significance analysis.

As regards the comparison between CBR-SPP and DL-SPP.sbert, we performed a one-sample *t*-test. As regards the comparison between DL-SPP.sbert and DL-SPP.bert, we performed a two-sample *t*-test. In all the tests, we used $\alpha = 0.05$ as level of significance. Whenever the *p*-value is lower than the level of significance, we can reject the statistical hypothesis of equivalence. Otherwise, no statistical differences can be identified. Table B.10 reports the numerical results.

As regards the comparison between DL-SPP.sbert and CBR-SPP, the *p*-values confirm that the improvements of the proposed approach are statistically significant across both datasets. The 95 % confidence intervals of DL-SPP.sbert do not overlap with CBR-SPP scores.

As regards the comparison between DL-SPP.sbert and DL-SPP.bert, the obtained *p*-values suggest that the differences between approaches are statistically significant for most *K* values. On *Dataset A*, DL-SPP.sbert outperforms DL-SPP.bert for small *K* values (up to *K* = 15), demonstrating a stronger ability to prioritize the most relevant spare parts within shorter recommendation lists. As *K* increases, the two models progressively converge toward similar sets of candidate parts, and their performances align. Conversely, on *Dataset B*, the advantage of DL-SPP.sbert is larger and consistent across all *K* values, confirming its superior generalization in a broader and more heterogeneous industrial context.

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